Q1 - Single Correct

The area bounded by the curve $y=\int_{1/8}^{\sin^2 x}\left(\sin^{-1}\sqrt{t}\right)dt+\int_{1/8}^{\cos^2 x}\left(\cos^{-1}\sqrt{t}\right)dt, 0\leq x\leq \pi/2$ and the curve satisfying the equation $y\left(x+y^3\right)dx=x\left(y^3-x\right)dy$ by passing through $(4,-2)$ is

(1) $\frac{7}{8}\left(\frac{3\pi}{16}\right)^2$

(2) $\frac{1}{8}\left(\frac{3\pi}{16}\right)^3$

(3) $\frac{1}{8}\left(\frac{3\pi}{16}\right)^4$

(4) None of these

Q2 - Single Correct

If $\lim_{x\to a}\frac{\int_a^t f(x)dx-\left(\frac{t-a}{2}\right)\{f(t)+f(a)\}}{(t-a)^3}=3a$ and $f(1)=-6, f'(1)=42$, then $f(0)$ is

(1) -18

(2) -60

(3) -80

(4) 20

Q3 - Single Correct

If $9+f''(x)+f'(x)=x^2+f^2(x)$ is the differential equation of a curve and P is a point of maxima of this curve, then the number of tangents which can be drawn from P to the circle $x^2+y^2=9$ is

(1) 2

(2) 1

(3) 0

(4) None of these

Q4 - Single Correct

Water is drained from a vertical cylinderical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y, where the constant of proportionality $k > 0$ depends on the acceleration due to gravity and the geometry of the hole. If t is measured in minutes and $k = 1/15$, then the time to drain the tank, if the water is $4 \mathrm{~m}$ deep to start with, is

(1) 30 min

(2) 45 min

(3) 60 min

(4) 80 min

Q5 - Single Correct

If the differential equation of the family of curve given by $y = Ax + Be^{2x}$, where A and B are arbitrary constant is of the form $(1-2x) \cdot \frac{d}{dx}\left(\frac{dy}{dx} + ly\right) + k\left(\frac{dy}{dx} + ly\right) = 0$, then the ordered pair (k, l) is

(1) $(2, -2)$

(2) $(-2, 2)$

(3) $(2, 2)$

(4) $(-2, -2)$

Q6 - Single Correct

A function $y = f(x)$ satisfies the condition $f'(x) \sin x + f(x) \cos x = 1$, $f(x)$ being bounded when $x \rightarrow 0$. If $I = \int_0^{\pi/2} f(x) dx$, then

(1) $\pi/2 < I < \pi^2/4$

(2) $\pi/4 < I < \pi^2/2$

(3) $1 < I < \pi/2$

(4) $0 < I < 1$

Q7 - Single Correct

A curve is such that the area of the region bounded by the coordinate axes, the curve and the ordiante of any point on it is equal to the cube of that ordinate. The curve represents

(1) a pair of straight lines

(2) a circle

(3) a parabola

(4) an ellipse

Q8 - Single Correct

The real value of m for which the substitution, $y=u^m$ will transform the differential equation, $2x^4y\frac{dy}{dx}+y^4=4x^6$ into a homogenous equation is

(1) $m=0$

(2) $m=1$

(3) $m=3/2$

(4) no value of m

Q9 - Single Correct

A curve C passes through origin and has the property that at each point (x, y) on it the normal line at that point passes through $(1,0)$. The equation of a common tangent to the curve C and the parabola $y^2=4x$ is

(1) $x=0$

(2) $y=0$

(3) $y=x+1$

(4) $x+y+1=0$

Q10 - Single Correct

A function $y=f(x)$ satisfies $(x+1)f'(x)-2\left(x^2+x\right)f(x)=\frac{e^{x^2}}{x+1}, \forall x>-1$. If $f(0)=5$; then $f(x)$ is

(1) $\left(\frac{3x+5}{x+1}\right)e^{x^2}$

(2) $\left(\frac{6x+5}{x+1}\right)e^{x^2}$

(3) $\left(\frac{6x+5}{(x+1)^2}\right)e^{x^2}$

(4) $\left(\frac{5-6x}{x+1}\right)e^{x^2}$

Q11 - Single Correct

The curve with the property that the projection of the ordinate on the normal is constant and has a length equal to a, is

(1) $a \log\left(\sqrt{y^2-a^2}+y\right)=x+C$

(2) $x+\sqrt{a^2-y^2}=c$

(3) $(y-a)^2=Cx$

(4) $ay=\tan^{-1}(x+C)$

Q12 - Single Correct

Consider the differential equation $dy/dx+y\tan x=x\tan x+1$. Then,

(1) the integral curves satisfying the differential equation and given by $y=x+C\sin x$

(2) the angle at which the integral curves cut the Y -axis is $\pi/2$

(3) tangents to all the integral curves at their point of intersections with Y -axis are parallel

(4) None of the above

Q13 - Single Correct

The subsitution $y=z^\alpha$ transforms the differential equation $\left(x^2y^2-1\right)dy+2xy^3dx=0$ into a homogenous differential equation for

(1) $\alpha=-1$

(2) $\alpha=0$

(3) $\alpha=1$

(4) no value of α.

Q14 - Single Correct

A curve passing through $(2,3)$ and satisfying the differential equation $\int_0^x t\cdot y(t)dt=x^2y(x),(x>0)$ is

(1) $xy=6$

(2) $xy=4$

(3) $x^2+y^2=13$

(4) $y^2=\frac{9}{2}x$

Q15 - Multiple Correct

A curve $y=f(x)$ has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of the point P from the X -axis. Then, the differential equation of the curve

(1) is homogenous

(2) can be converted into linear differential equation with some suitable substitution

(3) is the family of circles touching the X -axis at origin

(4) is the family of circles touching the Y -axis at origin

Q16 - Multiple Correct

The function $f(x)$ satisfying the equation $f^2(x)+4f'(x)f(x)+[f'(x)]^2=0$ is

(1) $f(x)=C\cdot e^{(2-\sqrt{3})x}$

(2) $f(x)=C\cdot e^{(2+\sqrt{3})x}$

(3) $f(x)=C\cdot e^{(\sqrt{3}-2)x}$

(4) $f(x)=C\cdot e^{-(2+\sqrt{3})x}$

Q17 - Multiple Correct

A function $y=f(x)$ satisfying the differential equation $\frac{dy}{dx}\sin x-y\cos x+\frac{\sin^2 x}{x^2}=0$ is such that, $y\to 0$ as $x\to\infty$, then which of the statements is/are correct?

(1) $\lim_{x\to 0} f(x)=1$

(2) $\int_0^{\pi/2} f(x)dx<3/2$

(3) $\int_0^{\pi/2} f(x)dx>1$

(4) $f(x)$ is odd function

Q18 - Multiple Correct

A differential function satisfies $f(x)=\int_0^x[f(t)\cos t-\cos(t-x)]dt$, then which of the following hold good?

(1) $\min f(x)=1-e$

(2) $\max f(x)=1-e^{-1}$

(3) $f''(\pi/2)=\mathrm{e}$

(4) $f'(0)=1$

Q19 - Multiple Correct

Let $dy/dx+y=f(x)$, where y is a continuous function of x with $y(0)=1$ and
$f(x)=\begin{cases} e^{-x}, & \text{if } 0\leq x<2 \\ e^{-2}, & \text{if } x>2 \end{cases}$, then which of the following hold good?

(1) $y(1)=2e^{-1}$

(2) $y'(1)=-e^{-1}$

(3) $y(3)=-2e^{-3}$

(4) $y'(3)=-2e^{-3}$

Q20 - Multiple Correct

The solution of $\frac{dy}{dx}=\frac{ax+h}{by+k}$ represents a parabola, if

(1) $a=-2, b=0$

(2) $a=-2, b=2$

(3) $a=0, b=2$

(4) $a=0, b=0$

Q21 - Multiple Correct

A normal is drawn at a point $P(x,y)$ of a curve. It meets the X -axis and the Y -axis in points A and B respectively, such that $\frac{1}{OA}+\frac{1}{OB}=1$, where O is origin, the equation of such a curve is a circle which passes through $(5,4)$ and has

(1) centre as $(1,1)$

(2) centre as $(2,1)$

(3) radius 5

(4) radius 4

Q22 - Multiple Correct

The graph of function $y=f(x)$ passing through the point $(0,1)$ and satisfying the differential equation $\frac{dy}{dx}+y\cos x=\cos x$, is such that

(1) it is a constant function

(2) it is periodic function

(3) it is neither an even nor an odd function

(4) it is continuous and differentiable function for all x

Q23 - Multiple Correct

If $y(x)$ satisfies the differential equation $y'+y=\cos x-\sin x$ and $y(\pi/2)=0$, then

(1) $y(0)=1$

(2) $y\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$

(3) $y\left(\frac{\pi}{3}\right)=2$

(4) $y\left(\frac{\pi}{4}\right)=\sqrt{2}$

Q24 - Multiple Correct

If $y(x)$ satisfy the differential equation $y'+y\tan x=2x+x^2\tan x$ and $y(0)=0$, then

(1) $y\left(\frac{\pi}{4}\right)=\frac{\pi^2}{16}$

(2) $y\left(\frac{\pi}{4}\right)=\sqrt{2}\frac{\pi^2}{16}$

(3) $y\left(\frac{\pi}{3}\right)=\frac{\pi^2}{9}$

(4) $y\left(\frac{\pi}{3}\right)=\sqrt{3}\frac{\pi^2}{9}$

Q25 - Multiple Correct

If $y(x)$ satisfy the differential equation $y' + \frac{y}{x} = \cos x + \frac{\sin x}{x}$ and $y(0) = 0$, then

(1) $y\left(\frac{\pi}{2}\right) = \frac{1}{2}$
(2) $y\left(\frac{\pi}{2}\right) = 1$
(3) $y\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$
(4) $y\left(\frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$

Q26 - Multiple Correct

If $y(x)$ satisfy the differential equation $y - \left(x + 2y^2\right) \cdot y' = 0$ and $y(0) = 1$, then x is

(1) $2y(y-1)$
(2) $y(y-1)$
(3) 4, when $y = 2$
(4) 1, when $y = 2$

Q27 - Multiple Correct

If $y(x)$ satisfy the differential equation $\frac{xdx+ydy}{xdy-ydx} = \sqrt{\frac{1-x^2-y^2}{x^2+y^2}}$ and $y(1) = 0$, then

(1) $y(0) = 0$
(2) $y(0) = -1$
(3) $y(0) = 2$
(4) $y(0) = -2$

Q28 - Multiple Correct

A tangent drawn to the curve $y = f(x)$ at $P(x, y)$ cut the X -axis and Y -axis at A and B respectively such that $BP : AP = 3 : 1$, given that $f(1) = 1$, then

(1) equation of the curve is $x\frac{dy}{dx} - 3y = 0$
(2) normal at $(1, 1)$ is $x + 3y = 4$

(3) curve passes through $(2, 1/8)$

(4) equation of the curve is $x\frac{dy}{dx} + 3y = 0$

Q29 - Paragraph 1

Passage I (For Question 29, 30)

Let $y = f(x)$ satisfies the equation $f(x) = (e^{-x} + e^{x})\cos x - 2x - \int_0^x (x-t) f'(t) dt$.

y satisfies the differential equation

(1) $\frac{dy}{dx} = y + e^{x} \cdot (\cos x - \sin x) - e^{-x}(\cos x + \sin x)$

(2) $\frac{dy}{dx} + y = e^{x} \cdot (\cos x - \sin x) - e^{-x}(\cos x + \sin x)$

(3) $\frac{dy}{dx} + y = e^{x}(\cos x + \sin x) - e^{-x}(\cos x - \sin x)$

(4) None of the above

Q30 - Paragraph 1

The value of $f'(0) + f''(0)$ equals

(1) -1

(2) 2

(3) 1

(4) 0

Q31 - Paragraph 2

Passage II (For Question 31, 32)

A curve $y = f(x)$ satisfies the differential equation $\left(1 + x^2\right) dy/dx + 2xy = 4x^2$ and passes through the origin.

The function $y = f(x)$ is

(1) strictly increasing, $\forall x \in R$

(2) such that it has a minima but no maxima

(3) such that it has a maxima but no minima

(4) None of the above

Q32 - Paragraph 2

The area enclosed by $y=f^{-1}(x)$, the X -axis and the ordinate at $x=2/3$ is

(1) $2\log 2$

(2) $\frac{4}{3}\log 2$

(3) $\frac{2}{3}\log 2$

(4) $\frac{1}{3}\log 2$

Q33 - Paragraph 3

Passage III (For Question 33, 34)

Differential equation $dy/dx=f(x)\cdot g(y)$ can be solved by seperating variable $\frac{dy}{g(y)}=f(x)dx$

The equation of curve to the point $(1,0)$ which satisfy the differential equation $\left(1+y^2\right)dx-xydy=0$ is

(1) $x^2+y^2=1$

(2) $x^2-y^2=1$

(3) $x^2+y^2=2$

(4) $x^2-y^2=2$

Q34 - Paragraph 3

The solution of the differential equation $\frac{dy}{dx}+\frac{1+y^2}{\sqrt{1-x^2}}=0$ is

(1) $\tan^{-1}y+\sin^{-1}x=C$

(2) $\tan^{-1}x+\sin^{-1}y=C$

(3) $\tan^{-1}y\cdot\sin^{-1}x=C$

(4) $\tan^{-1}y-\sin^{-1}x=C$

Q35 - Paragraph 4

Passage IV (For Question 35, 36)

The slope of tangent to a curve at any point (x, y) on it is given by $\frac{y}{x} - \cot\left(\frac{y}{x}\right) \cdot \cos\left(\frac{y}{x}\right)(x > 0, y > 0)$ and solution of curve is given by

$\sec\left(\frac{y}{x}\right) = f(x) + C$

The value of $f(x)$ is

(1) $\log|x|$

(2) $\log|2x|$

(3) $\log\left|\frac{1}{x}\right|$

(4) None of these

Q36 - Paragraph 4

The value of $\lim_{x \to 0} \frac{f(1+x)}{x}$ is

(1) -1

(2) 1

(3) $\frac{1}{2}$

(4) 2

Q37 - Paragraph 5

Passage V (For Question 37, 38)

Let $f : [0, 1] \to R$ be a function. Suppose the function f is twice differentiable; $f(0) = f(1) = 0$ and satisfies $f''(x) - 2f'(x) + f(x) \geq e^x, x \in [0, 1]$.

If the function $e^{-x} f(x)$ assumes its minimum in the interval $[0, 1]$ at $x = 1/4$, then which of the following is true?

(1) $f(x) < f(x), \frac{1}{4} < x < \frac{3}{4}$

(2) $f'(x) > f(x), 0 < x < \frac{1}{4}$

(3) $f'(x) < f(x), 0 < x < \frac{1}{4}$

(4) $f'(x) < f(x), \frac{3}{4} < x < 1$

Q38 - Paragraph 5

Which of the following is true for $0 < x < 1$?

(1) $0 < f(x) < \infty$

(2) $-\frac{1}{2} < f(x) < \frac{1}{2}$

(3) $-\frac{1}{4} < f(x) < 1$

(4) $-\infty < f(x) < 0$

Q39 - Integer Type

The solution of $y' + \frac{y}{x} = \frac{1}{(1+\log xy)^2}$ is $xy\left[1 + (\log xy)^2\right] = \frac{x^\lambda}{\mu} + C$, then the value of $(\lambda + \mu)$ is

Q40 - Integer Type

The solution of $y'\left(x^2 - 2x^3y^3\right) = \left(3x^2y^4 + 2xy\right)$ is $x^\lambda y^2 + \frac{x^\mu}{y} = C$, then the value of $(\lambda + \mu)$ is

Q41 - Integer Type

The solution of $y(2xy+1)dx + x\left(1 + 2xy + x^2y^2\right)dy = 0$ is $-\frac{1}{2(xy)^2} - \frac{\lambda}{xy} + \log y = C$, then the value of λ is

Q42 - Integer Type

The solution of $\cos\left(\frac{x}{y}\right)(ydx - xdy) = xy^3(xdy + ydx)$ is $\sin\left(\frac{x}{y}\right) = \frac{(xy)^\lambda}{\mu} + C$, then the value of $(\lambda + \mu)$ is

Q43 - Integer Type

The solution of $\left\{y\left(1 + \frac{1}{x}\right) + \cos y\right\}dx + \{x + \log x - x\sin y\}dy = 0$ is $xy + y\log x + x^\lambda \cos y = C$, then the value of λ is

Q44 - Integer Type

The solution of $xy' - y = mx^2\sqrt{x^2 - y^2}$ is $\sin^{-1}\left(\frac{y}{x}\right) = m\frac{x^\lambda}{\mu} + C$, then the value of $(\lambda - \mu)$ is

Q45 - Integer Type

If the solution of $\frac{dy}{dx} = (x + y - 1) + \frac{x+y}{\log(x+y)}$ is $1 + \log(x + y) - \log\{k + \log(x + y)\} = x + C$, then k is equal to

Q46 - Integer Type

Let f be a real valued differentiable function on R (the set of all real numbers) such that $f(1) = 1$. If the y - intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P, then the value of $f(-3)$ is

Q47 - Integer Type

If the solution of $\frac{dy}{dx} = \frac{\sin y + x}{\sin 2y - x\cos y}$ is $\sin^2 y = x\sin y + \frac{x^2}{a} + C$, then the value of a is

Q48 - Integer Type

If the solution of $\left(1 + 2e^{x/y}\right)dx + 2e^{x/y}\cdot\left(1 - \frac{x}{y}\right)dy = 0$ is given by $x + 2y[f(x)]^{1/y} = C$, then $\lim_{x\to 0}\frac{f(x)-1}{x}$ is equal to

Q49 - Integer Type

Let $f : [1, \infty) \to [2, \infty)$ be a differentiable function such that $f(1) = 2$. If $6\int_1^x f(t)dt = 3xf(x) - x^3$ for all $x > 1$, then the value of $f(2)$ is

Q50 - Integer Type

A function $f(x)$ satisfying $\int_0^1 f(tx)dt = nf(x)$, where $x > 0$ is $f(x) = cx^{(1-\lambda)/\mu}$, then the value of $(\lambda + \mu)$ is

Answer Key

Q1 (3)	**Q2** (2)	**Q3** (3)	**Q4** (3)
Q5 (1)	**Q6** (1)	**Q7** (3)	**Q8** (3)
Q9 (1)	**Q10** (2)	**Q11** (1)	**Q12** (3)
Q13 (1)	**Q14** (1)	**Q15** (1, 2, 4)	**Q16** (3, 4)
Q17 (1, 2, 3)	**Q18** (1, 2, 3)	**Q19** (1, 2, 4)	**Q20** (1, 3)
Q21 (1, 3)	**Q22** (1, 2, 4)	**Q23** (1, 2)	**Q24** (1, 3)
Q25 (2, 3)	**Q26** (1, 3)	**Q27** (1)	**Q28** (3, 4)
Q29 (2)	**Q30** (4)	**Q31** (1)	**Q32** (3)
Q33 (2)	**Q34** (1)	**Q35** (3)	**Q36** (1)
Q37 (3)	**Q38** (4)	**Q39** (4)	**Q40** (5)
Q41 (2)	**Q42** (4)	**Q43** (1)	**Q44** (0)
Q45 (1)	**Q46** (9)	**Q47** (2)	**Q48** (1)
Q49 (6)	**Q50** (1)		